SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	01 July 2026
Team ID	
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to improve the **credit card approval process** by using **Machine Learning** techniques to predict whether a credit card application should be approved or rejected. The system reduces manual effort, improves prediction accuracy, and provides instant decision support through a user-friendly web application.

Project Overview	
Objective	The primary objective of this project is to develop a machine learning-based system that predicts credit card approval decisions quickly and accurately using applicant information such as income, employment, education, family status, and other relevant features.
Scope	The project focuses on analyzing credit card applicant data, preprocessing the dataset, training multiple machine learning models, selecting the best-performing model, and deploying it as a Flask web application for real-time prediction..
Problem Statement	
Description	The traditional credit card approval process is time-consuming and requires manual verification of applicant information. This may result in delays, inconsistent decisions, and increased operational costs.

Impact	Implementing an automated machine learning-based credit card approval system will: • Improve decision-making accuracy. • Reduce manual processing time. • Minimize operational costs. • Enhance customer satisfaction through faster responses. • Support banks in making consistent approval decisions.
Proposed Solution	
Approach	The proposed solution uses a **Machine Learning classification model (Logistic Regression)** to analyze applicant information and predict whether a credit card application should be approved or rejected. The trained model is integrated with a Flask web application, enabling users to receive instant predictions through an interactive interface.
Key Features	Machine learning-based credit card approval prediction. User-friendly web interface developed using HTML, CSS, and Flask. Real-time prediction based on applicant details.
	- Data preprocessing and feature engineering for improved accuracy. - Model deployment on Render for online accessibility.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU

Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv